Web-Based Interactive CCU Orientation Manual: Reducing Arterial Line-Related Compartment Syndrome Through Digital Education

Mu-Yang Hsieh MD, PhD, FESC[1,2], Chen-Wei Lien MD[1], Ru-Ying Hsu MD[1], Li-Ta Keng MD[1]

[1]Division of Cardiology, National Taiwan University Hospital Hsinchu Branch
[2]Graduate Institute of Clinical Medicine, National Taiwan University

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Background & Problem

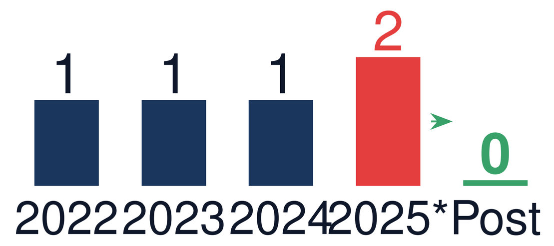
The Challenge

Acute myocardial infarction (AMI) patients in the coronary care unit (CCU) require intensive hemodynamic monitoring. Arterial line placement is a common procedure, but patients on **dual antiplatelet therapy (DAPT)** combined with **anticoagulation** face elevated bleeding risks.

Critical Issue: Brachial artery puncture in anticoagulated patients can lead to **compartment syndrome**, requiring emergency fasciotomy.

Incidence Data (2022-2025)

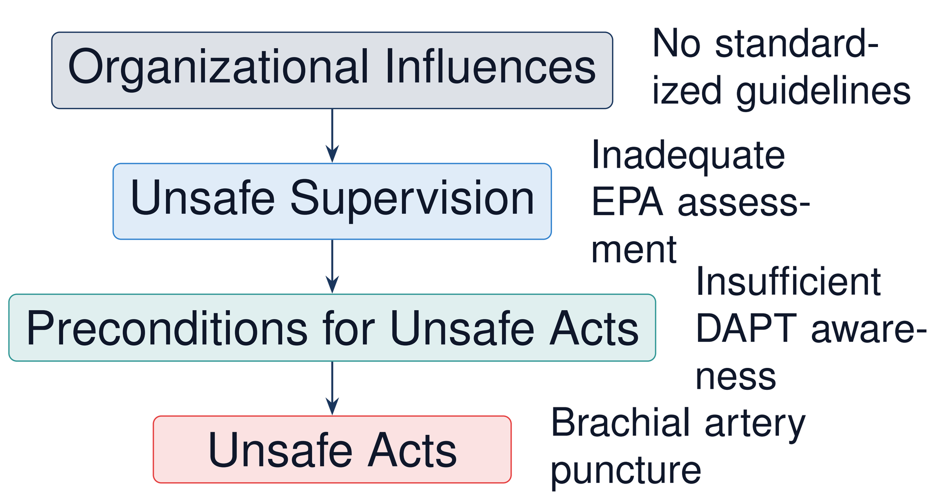
Compartment Syndrome Cases



*Index case July 2025: 67-year-old male with AMI, multiple brachial artery puncture attempts while on Brilinta + Clexane + Heparin

Methods: HFACS Analysis

We applied the **Human Factors Analysis and Classification System (HFACS)** framework to identify root causes and design educational interventions.



Root Causes Identified

1. Lack of standardized arterial line placement guidelines
2. No risk stratification for anticoagulated patients
3. Insufficient education on DAPT bleeding complications
4. Missing EPA-based competency assessment

Study Design

Intervention Period: August 2025 – Present
Setting: 10-bed CCU, tertiary medical center
Target Learners: Residents, Nurse Practitioners

Educational Framework:

- Root Cause Analysis-driven curriculum
- Competency-Based Medical Education (CBME)
- Entrustable Professional Activities (EPA)
- Just-in-time digital learning resources

The Intervention: Interactive Web-Based Manual

CCU Orientation Manual v2.1

React-based Single-Page Application

9 Interactive Modules

1. Daily Checklist
2. Safety Cases
3. ACS Orders
4. HF Certification
5. Special Care
6. Drug Reference
7. Emergency Protocols
8. Chart Writing Guide
9. Self-Assessment Quiz

Key Educational Features:

Safety Case Studies

Interactive Checklists

6P Monitoring Protocol

Self-Assessment Quizzes
10-question knowledge checks with immediate feedback

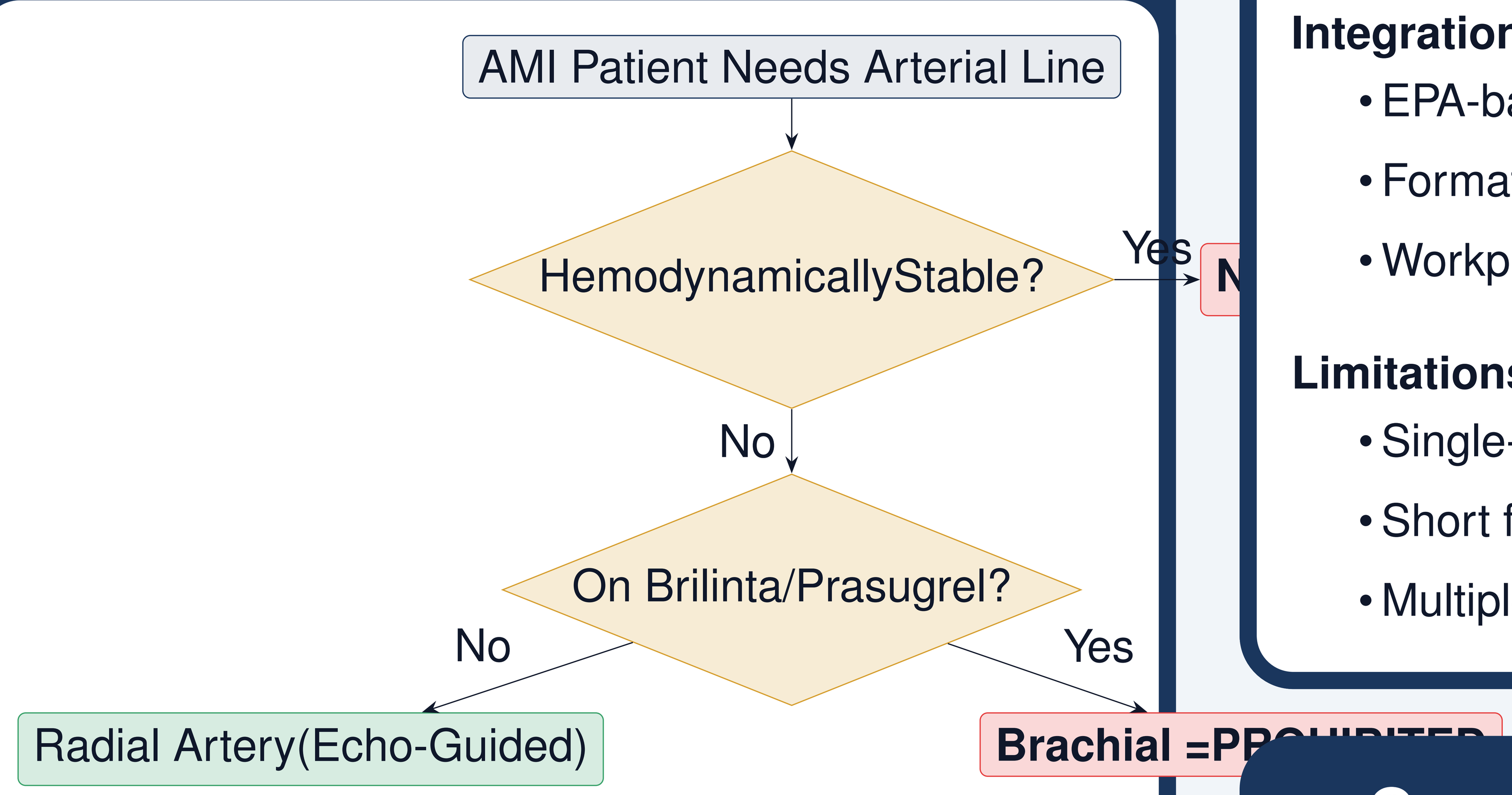
Critical Safety Message Embedded:

"If patient is on Brilinta/Prasugrel + anticoagulation, **DO NOT** perform brachial artery puncture"

Scan to Access the Manual:

ntuh-hsinchu-cv.github.io/ccu-orientation-manual.html

Arterial Line Placement Algorithm



Post-removal monitoring: 6P assessment every 15 minutes for 1 hour before ICU discharge

Results

Primary Outcome

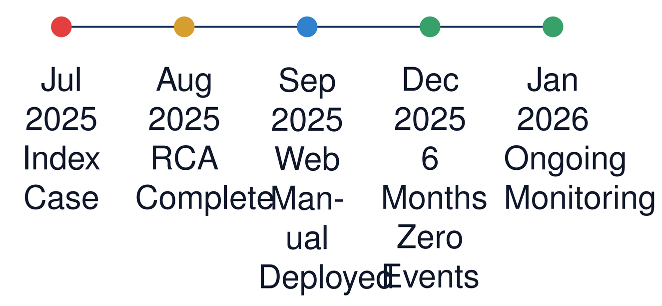
BEFORE 1.6 cases/year → AFTER 0 cases in 6 months

100% Reduction
in arterial line-related compartment syndrome incidents

Secondary Outcomes

Metric	Before	After
Radial artery first-choice rate	65%	95%
Echo-guided puncture rate	40%	90%
6P monitoring compliance	—	100%
Quiz pass rate (≥80%)	—	92%

Implementation Timeline



Discussion

Why Web-Based Education Works:

- **Accessibility:** 24/7 mobile-friendly access
- **Interactivity:** Active learning with quizzes
- **Just-in-time:** Available at point of care
- **Updateable:** Rapid iteration based on new cases

Integration with CBME Framework:

- EPA-based competency milestones
- Formative assessment via self-quizzes
- Workplace-based learning resources

Limitations:

- Single-center study
- Short follow-up period (6 months)
- Multiple concurrent interventions

Conclusions & Take-Home Messages

1. HFACS-driven RCA enables sys-

2. Web-based in-

3. Digital educa-

4. Integration with EPA framework supports